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Jawaban

1. Turunan implisit ($y' = f(x, y)$)

a. $(3xy + 7)^2 = 6y$

$$\Rightarrow 2(3xy + 7) \left(\frac{d}{dx} 3xy + 7 \right) = 6y'$$

$$\Rightarrow 2(3xy + 7)(3(y + y'x)) = 6y'$$

$$\Rightarrow y' = (3xy + 7)(y + y'x)$$

$$= 3xy^2 + 7y + 3x^2y'y + 7y'x$$

$$\Rightarrow y' - 3x^2y'y - 7y'x = 3xy^2 + 7y$$

$$\Rightarrow y'(1 - 3x^2y - 7x) = 3xy^2 + 7y$$

$$\Rightarrow \frac{\partial y}{\partial x} = \frac{3xy^2 + 7y}{1 - 3x^2y - 7x}$$

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b. $x + \tan(xy) = 0$

$$\Rightarrow 1 + (y + y'x) \sec^2(xy) = 0$$

$$\Rightarrow y \sec^2(xy) + xy' \sec^2(xy) = -1$$

$$\Rightarrow xy' \sec^2(xy) = -1 - y \sec^2(xy)$$

$$\Rightarrow \frac{\partial y}{\partial x} = - \frac{(1 + y \sec^2(xy))}{x \sec^2(xy)}$$

$$= -\frac{1}{x} \cos^2(xy) - \frac{y}{x}$$

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2 Turunan implisit dua kali ($y'' = f(x, y)$)

a $y^2 - 2x = 1 - 2y$

$\Rightarrow 2yy' - 2 = -2y'$

$\Rightarrow 2yy' + 2y' = 2 \Rightarrow yy' + y' = 1$

$\Rightarrow y'(1+y) = 1$

$\Rightarrow \frac{2y}{2x} = \frac{1}{1+y} \quad (1)$

Kemudian, cari turunan dari $\frac{\partial y}{\partial x}$ terhadap x

$\frac{\partial}{\partial x} \frac{\partial y}{\partial x}$

$\Rightarrow -\frac{y'}{(1+y)^2}$, Substitusi y' dengan (1)

$\Rightarrow \boxed{\frac{d^2 y}{dx^2} = -\frac{1}{(1+y)^3}}$

b $2\sqrt{y} = x - y$

$\Rightarrow \frac{y'}{\sqrt{y}} = 1 - y' \Rightarrow y' = (1 - y')\sqrt{y}$

$\Rightarrow y' = \sqrt{y} - y'\sqrt{y}$

$\Rightarrow y' + y'\sqrt{y} = \sqrt{y}$

$\Rightarrow y'(1 + \sqrt{y}) = \sqrt{y}$

$\Rightarrow y' = \frac{\sqrt{y}}{1 + \sqrt{y}} = 1 - \frac{1}{1 + \sqrt{y}}$

$\Rightarrow \frac{\partial}{\partial x} y' = \frac{\left(\frac{\partial}{\partial x} 1 + \sqrt{y}\right)}{(1 + \sqrt{y})^2}$

$= \frac{\frac{1}{2\sqrt{y}} \cdot y'}{1 + y + 2\sqrt{y}} = \frac{\frac{\sqrt{y}}{1 + \sqrt{y}}}{2\sqrt{y} \cdot (1 + \sqrt{y})^2}$

$= \boxed{\frac{1}{2} \cdot \frac{1}{(1 + \sqrt{y})^3}}$

3. Pers $y = mx + b$ $(1, -1)$ dari

$$(x^2 + y^2)^2 = (x - y)^2 \Rightarrow (x^2 + y^2)^2 = x^2 + y^2 - 2yx$$

$$\Rightarrow 2(x^2 + y^2)(2x + 2yy') = 2x + 2yy' - 2(y + y'x)$$

Malas mencari bentuk tertutup, substitusi:

$$x = 1, y = -1, y' = m$$

$$\Rightarrow 2 \cdot 2 \cdot (2 + -2m) = 2 - 2m - 2(-1 + m)$$

$$\Rightarrow 8 - 8m = 2 - 2m + 2 - 2m$$

$$\Rightarrow 4 = 4m$$

$$\Rightarrow m = 1$$

$$y = x + b \Rightarrow -1 = 1 + b \Rightarrow b = -2$$

Dengan demikian,

$$\boxed{y = x - 2}$$

4. $xy + y^2 = 1$ di $(0, 1)$

$$\Rightarrow y + xy' + 2yy' = 0$$

$$\Rightarrow xy' + 2yy' = -y$$

$$\Rightarrow y'(x + 2y) = -y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y}{x + 2y}, \text{ nilai } y'(0, 1) = -\frac{1}{2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{-y'(x + 2y) - y(1 + 2y')}{(x + 2y)^2}$$

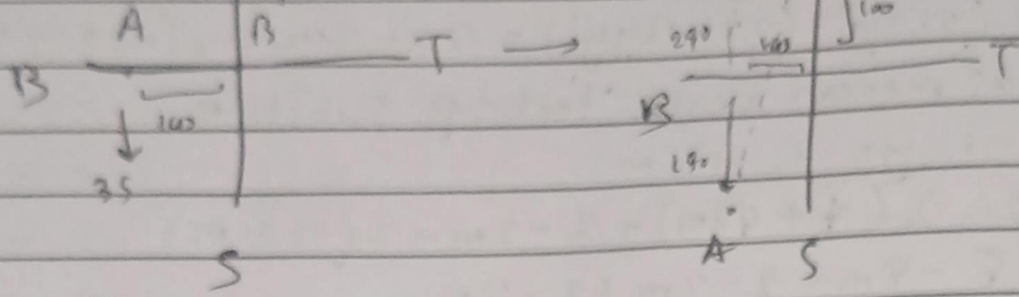
$$= \frac{-(-\frac{1}{2})2(-1) - (-1)(1 + 2(-\frac{1}{2}))}{(-1)^2}$$

$$= \boxed{-\frac{1}{4}}$$

[- (min) di bawah
itu dari faktor (-1) ,
bukan hasil kuadrat]

25 32x

25x92 (00)



$$|\vec{AB}|_{12.00} = 100$$

$$|\vec{AB}|_{1670} = 260$$

Rumus jarak $\rightarrow z^2 = x^2 + y^2$

Karena hanya ada perpindahan pada sumbu-y, misal
total jarak $= y = a + b$ dengan $a = A$ (jarak), a' kecepatan
 $b = B$ (jarak), b' kecepatan

$$z^2 = x^2 + (a+b)^2 \text{ terhadap waktu } (t) \text{ dalam jam}$$

$$\Rightarrow 2zz' = 2xx' + 2(a+b)(a'+b')$$

$$\Rightarrow z' = \frac{(a+b)(a'+b')}{z}$$

$$= 240 \cdot (60)$$

260

$$= \frac{710}{13}$$

$$\approx 55,389615 \text{ km/Jam}$$

6 $y = 2 \sin\left(\frac{\pi x}{2}\right)$

Pakai rumus jarak dengan $x = \frac{1}{3}$, $y = 1$

$$z^2 = x^2 + y^2 \Rightarrow z = \sqrt{\left(\frac{1}{3}\right)^2 + 1^2} = \frac{1}{3}\sqrt{10}$$

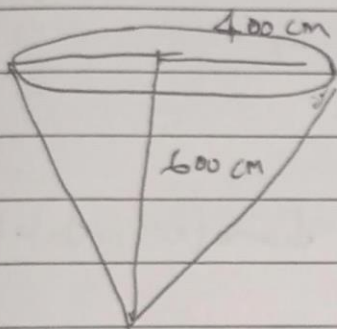
$$= x^2 + 4 \sin^2\left(\frac{\pi x}{2}\right)$$

$$\Rightarrow 2z z' = 2x x' + \frac{\pi}{2} \cdot 4 \cdot 2 \sin\left(\frac{\pi x}{2}\right) \cdot \cos\left(\frac{\pi x}{2}\right) \cdot x'$$

$$\Rightarrow z' = \frac{x x' + 4 \pi \sin\left(\frac{\pi x}{2}\right) \cos\left(\frac{\pi x}{2}\right) \cdot x'}{2z}$$

$$= \frac{\frac{1}{3} \cdot \sqrt{10} + 4 \pi \cdot \frac{1}{2} \cdot \frac{1}{2} \sqrt{3} \cdot \sqrt{10}}{\frac{1}{3}\sqrt{10}}$$

$$= \boxed{1 + \frac{3\pi\sqrt{3}}{2} \text{ cm/s}}$$



$$1000 \text{ cm}^3/\text{m}$$

$$\Delta h = 20 \text{ cm/menit}$$

$$V' = V'_{\text{masuk}} - V'_{\text{keluar}} \quad (1)$$

$$V = \frac{1}{3} \pi r^2 h$$

Pakai kesebandingan segitiga, didapat

$$\frac{r}{h} = \frac{r_{\text{max}}}{h_{\text{max}}} = \frac{400}{600} = \frac{2}{3} \Rightarrow h = \frac{3}{2} r \Rightarrow r = \frac{2}{3} h$$

$$\Rightarrow V = \frac{1}{3} \pi \left(\frac{2}{3} h\right)^2 \cdot h$$

$$= \frac{\pi h^3}{27}$$

$$\Rightarrow V' = \frac{\pi h^2}{27} \cdot h' = \frac{\pi h^2}{27} \cdot 20 = \frac{20\pi h^2}{27}$$

$$\text{Maka, laju } V'(200) = \frac{20\pi h^2}{27} \Big|_{h=200}$$

$$= \frac{80000\pi}{27} \text{ cm}^3/\text{menit}$$

Pakai (1), didapat $V'_{\text{masuk}} = V' + V'_{\text{keluar}}$

$$= \boxed{1000 + \frac{80000\pi}{27} \text{ cm}^3/\text{menit}}$$

8

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2, \quad \frac{dS}{dt} = 1 \text{ cm}^2/\text{menit}, \quad S = \pi d^2$$

$$\frac{dS}{dt} = 8\pi r \cdot r' \quad (?)$$

$$\frac{dS}{dt} = 2\pi d d'$$

$$\Rightarrow r' = \frac{1}{8\pi r}$$

$$\Rightarrow d' = \frac{1}{2\pi d}$$

$$8\pi r$$

$$2\pi d$$

$$d = 10 \text{ cm} \Rightarrow r = 5$$

$$d = 10 \text{ cm}$$

$$\Rightarrow r'(5) = \frac{1}{8\pi(5)}$$

$$\frac{dS}{dr}$$

$$\Rightarrow d'(10) = \frac{1}{20\pi}$$

$$20\pi$$

$$= \frac{1}{40\pi} \cdot 2$$

$$\hookrightarrow d = 2r$$

$$40\pi$$

$$dd = 2dr$$

$$= \frac{1}{20\pi}$$

$$\Rightarrow \frac{dd}{dr} = 2$$

$$20\pi$$

$$dr$$

Dengan demikian, laju pertumbuhannya diameter adalah

$$\frac{1}{20\pi} \text{ cm/menit}$$

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